When Does GRPO Learn? A Small-Scale Empirical Study of Success Rate, Group Size, and Reward Sparsity in RLVR

CMPT 981 Project Proposal

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Abstract

Reinforcement learning with verifiable rewards (RLVR), and in particular Group Relative Policy Optimization (GRPO), is now the standard recipe for training reasoning in large language models. Recent theory makes concrete predictions about its dynamics: how many sampled groups carry a learning signal, whether GRPO amplifies the success probability, and when a step size is too large. These predictions are derived for exact gradients and idealized policies, and are hard to check at LLM scale, where every run is expensive and the true success probability is unknown. We propose an experiment-centred project that tests them on a small synthetic task with a binary verifier. Its difficulty is set exactly by the answer length, and the policy's success probability can be computed in closed form, so runs take seconds on a CPU. We vary the initial success rate, the number of sampled answers per prompt, reward sparsity, KL strength, and step size, and compare each result with the corresponding theoretical prediction. As a small theoretical contribution, we extend the success-amplification fixed point of Mroueh [2025] from infinite to finite group size, which gives closed forms for mean-only and leave-one-out baselines and a testable prediction of how the group size shifts GRPO's fixed point. We also show that on-policy SGD shares these fixed points with Mroueh's exact iteration but not their stability. We close with a confirmation on a 0.5B-parameter LLM.

1 Introduction and Motivation

Reinforcement learning (RL) learns a policy of actions from a reward signal that is often sparse and delayed. In the context of large language models (LLMs), RL is used to align the model's responses with human preferences or task-specific objectives, most notably through reinforcement learning from human feedback (RLHF) [Ouyang et al., 2022]. However, RLHF can be expensive and time-consuming, as it requires humans to label which responses they prefer, and those labels are used to train a reward model.

In 2025, DeepSeek-R1-Zero [Guo et al., 2025] showed that a language model could learn long chain-of-thought reasoning from a rule-based reward, mainly a binary check of whether the final answer is correct, without any human feedback. This approach is called reinforcement learning with verifiable rewards (RLVR). The model was trained using Group Relative Policy Optimization (GRPO) [Shao et al., 2024], which samples multiple answers per prompt and normalizes the advantages by the group standard deviation. Since then, practitioners have relied on rules of thumb, such as sampling more answers for problems with a low success rate, normalizing advantages by

the group metrics, and averaging the loss over tokens. However, each of these choices changes the fundamental optimization problem, and it is still unclear which ones matter and why.

Initial theoretical work provides precise answers in simplified settings [Mroueh, 2025; Suk and Duan, 2025; Thrampoulidis et al., 2026; Bay and Yearick, 2026]. Examples include the success amplification property of GRPO [Mroueh, 2025], which states that, with a KL penalty, the success probability settles at a fixed point higher than that of the reference model, and the gradient gap [Suk and Duan, 2025], the difference between the expected score of correct and incorrect responses, which is used to derive a step-size threshold below which training converges. However, these results are derived under simplified conditions, far from the stochastic, finite-sample regime in which GRPO is actually run. For example, they assume exact expectations and, in the case of Mroueh [2025], an exactly solved inner optimization problem. Testing them directly is also hard, because LLM experiments cannot isolate a single variable: the success rate, task difficulty, and response length all change together.

Our project aims to bridge the gap between theory and practice. We plan controlled experiments that will allow us to sweep a grid of parameters, including the initial success rate, group size, reward sparsity, KL strength, and step size. By changing one parameter at a time and observing the effect on the learning dynamics, we can test the theoretical predictions in a setting closer to practice, with finite samples, stochastic gradients, and a learned policy. These results will offer further insight into the dynamics of RLVR and GRPO in practical settings. This can lead to a better understanding of the optimization problem and help guide how these settings are chosen when training LLMs with verifiable rewards.

2 Related Work

GRPO and RLVR. GRPO [Shao et al., 2024] drops PPO's value network. For each prompt it samples a group of G answers and uses the group-normalized reward $(r_i - \bar{r})/\mathrm{std}(r)$ as the advantage. DeepSeek-R1 [Guo et al., 2025] scaled this up with rule-based verifiers (an accuracy reward plus a format reward), $G = 16$, and a small KL coefficient. Dr. GRPO [Liu et al., 2025] argues that the std-normalization and the per-response length normalization introduce biases, and RLOO [Ahmadian et al., 2024] uses a leave-one-out baseline instead. We use all three as advantage variants.

Success amplification [Mroueh, 2025]. For binary rewards with a reverse-KL penalty and no clipping, the paper shows that GRPO's objective is a contrastive loss between correct and incorrect samples from the previous policy. If each iteration solves the KL-regularized problem exactly, the success probability follows a recurrence $p_n = h(p_{n-1})$, and every fixed point satisfies $p^* > p_{\mathrm{ref}}$, so GRPO amplifies success. For mean-only normalization the fixed point is explicit; for mean+std normalization it has no closed form and need not be unique. A local-convergence condition gives a lower bound on β, and the paper reports divergent iterates at $\beta = 5$ with $p_{\mathrm{ref}} = 0.001$. We test whether SGD training reaches the predicted p^* across KL strengths, extend the prediction to finite G, and ask whether SGD inherits the iteration's instability (Section 3.1). Davis and Recht [2025] give a complementary view: with binary rewards, several RL algorithms are stochastic gradient ascent on a monotone transform of the success probability, and for GRPO the transform is $\arcsin\sqrt{p}$. Rad et al. [2026] analyse GRPO on a bandit as a replicator-style flow and show a phase transition under verifier noise; theirs is the closest analysis in style to ours, but it has no KL penalty. Ding [2026] list ways the KL term misbehaves in group policy optimization, including KL updates on groups whose rewards are all equal.

Gradient gap and step-size thresholds [Suk and Duan, 2025]. The paper defines the gradient gap, the difference between the expected score of correct and incorrect responses, and proves a sufficient step-size condition for convergence together with lower-bound constructions where a constant step makes performance decrease to zero. With exact gradients, the threshold scales as $1/T$ in the response length T and shrinks with the success rate through $(1 - J)$; the paper argues that GRPO's $1/|y|$ normalization provides exactly this scaling and Dr. GRPO's does not. Its experiments use a single step size. We test whether a sharp transition appears in a stochastic, learned-policy setting, and how it moves with response length when responses of different lengths share one learning rate. Two recent results sharpen this question. Ge et al. [2026] show that std-normalization acts as an adaptive step size matched to local curvature, with a speed-up that depends on how much the per-prompt reward variance differs across prompts. Ding et al. [2026] prove that with outcome rewards no length weighting is both unbiased and length-invariant; GRPO and Dr. GRPO are the two ends of the family $f_\alpha(L) = L^{\alpha-1}$, $\alpha \in [0, 1]$.

Group size and informative prompts. A group with all-equal rewards has zero advantage. DAPO's dynamic sampling [Yu et al., 2025] and selective rollouts [Zheng et al., 2025] discard such groups, and Foster et al. [2025] sample prompts in proportion to $p(1 - p)$. Bay and Yearick [2026] show that GRPO, Dr. GRPO and DAPO differ only in how they use the group standard deviation, derive a closed-form group-size law in $G\,p(1-p)$, and check it on a Bernoulli toy model at $G = 8$ and by subsampling logged rollouts. Ghosh et al. [2026] model the mean reward trajectory as a damped oscillator and predict that it is nearly independent of G to leading order. Zhou et al. [2026] show that the GRPO gradient is a U-statistic, derive its mean squared error, and propose a scaling law for the group size. Nie et al. [2026] prove that when the success rate varies across prompts, the fraction of all-equal groups exceeds the prediction from the average success rate, and that a fixed-reference advantage $A = 2r - 1$ outperforms Dr. GRPO at $G = 4$ on GSM8K. Empirical results pull in two directions: Hu et al. [2025] revive stalled training by scaling to hundreds of rollouts per prompt, while Wu et al. [2025] retain 97.6% of the performance of $G = 16$ with $G = 2$. Plyusov et al. [2026] show that a group can have mixed rewards yet miss rare correct solutions, with a non-monotonic dependence on G. Prompt selection methods choose prompts by predicted learnability: ThinkPrior starts from an offline difficulty prior [Sha et al., 2026], and ARCUS tracks difficulty in Fisher–Rao coordinates of the pass rate [Okonkwo et al., 2026].

We do not claim the informative-group fraction (Eq. (3)) or the $p(1 - p)$ scaling as new. The closest prior work to ours is Bay and Yearick [2026], Zhou et al. [2026], Nie et al. [2026] and Plyusov et al. [2026], each of which varies G in theory or in a toy model. Our contribution is a controlled *training* grid over G, p_0, KL strength and step size, with every quantity measured exactly, compared against these closed forms.

Pass@K advantage shaping. Walder and Karkhanis [2025] and Chen et al. [2025] train directly for pass@K. Thrampoulidis et al. [2026] show that advantage-shaping heuristics, such as hard-example upweighting, implicitly maximize surrogate rewards such as pass@K, and derive an unbiased leave-one-out pass@K advantage. We test whether optimizing pass@K preserves response diversity and at what cost to pass@1.

Entropy and diversity collapse. Yue et al. [2025] show that RLVR-trained models beat their base model at small k but lose to it at large k. Cui et al. [2025] trace the fall in entropy to the covariance between a token's log-probability and its advantage, and fit an empirical law $R = -ae^H + b$ between performance and entropy. Yuan et al. [2026] attribute diversity collapse to overtraining

prompts that are already solved, and find that updating only on prompts with zero observed successes raises pass@256 above the base model. Shao et al. [2025] find that even random rewards improve some models, which argues for a random-reward control in LLM experiments.

3 Problem Formulation

A synthetic verifiable task. A prompt is a feature vector $x \in \mathbb{R}^d$. A response is a sequence $y = (y_1, \ldots, y_L)$ of L tokens, each from a vocabulary of K options. A hidden rule assigns each position a correct token $c_t(x) = \arg\max_k (W_t x)_k$. The verifier returns

$$r(x, y) = \mathbf{1}\{\, y_t = c_t(x) \;\; \forall t = 1, \ldots, L \,\}, \tag{1}$$

so reward is all-or-nothing, as with exact-match answers or unit tests. A uniform policy succeeds with probability $p_0 = K^{-L}$, so the answer length L and vocabulary size K set the reward sparsity exactly. To vary the initial success rate without changing the task, we also warm-start the policy with a short supervised phase that stops at a target p_0. This plays the role of a base model. For the step-size experiment we also use a *mixed-length* variant in which each prompt has its own length $L(x) \in \{2, \ldots, 8\}$. For the diversity experiment we use a *multi-solution* variant in which m of the K tokens are correct at each position, so a prompt has m^L correct responses and diversity among correct answers can be measured; the base model starts with unequal probabilities on them, so that some correct responses are rare.

Policy. $\pi_\theta(y \mid x) = \prod_{t=1}^{L} \pi_\theta(y_t \mid x, t)$, where each factor is the softmax output of a small two-layer MLP. Because the policy factorizes, the exact success probability on a prompt is $p(x) = \prod_t \pi_\theta(c_t(x) \mid x, t)$, and pass@$k = 1 - (1 - p(x))^k$. Our learning curves therefore have no evaluation noise, and the KL to π_{ref} can be computed exactly rather than estimated. The factors are conditionally independent given (x, t), so the policy is not autoregressive: a length-L response is a product of L coupled bandits sharing one network. With $L = 1$ the policy is a single softmax, which matches the bandit setting used in the theory papers.

Objective and update. We maximize the KL-regularized expected reward

$$J(\theta) = \mathbb{E}_x \Big[\mathbb{E}_{y \sim \pi_\theta(\cdot|x)} r(x, y) - \beta \, \mathrm{KL}(\pi_\theta(\cdot \mid x) \,\|\, \pi_{\text{ref}}(\cdot \mid x)) \Big]. \tag{2}$$

Each step samples B prompts and G responses per prompt, computes advantages A_i within each group, and takes one gradient step on $-\frac{1}{BG} \sum A_i \, \ell_i(\theta) + \beta \, \mathrm{KL}$, where $\ell_i = \log \pi_\theta(y_i \mid x)$ is the summed log-probability unless stated otherwise. Training is fully on-policy, so PPO's ratio clipping is inactive. We compare four advantages: `grpo` $(r_i - \bar{r})/\sqrt{\bar{r}(1-\bar{r}) + \epsilon}$; `mean-only` $r_i - \bar{r}$ (Dr. GRPO); `rloo`, which subtracts the leave-one-out mean; and the unbiased pass@k advantage of Thrampoulidis et al. [2026], implemented in its combinatorial form to avoid the $0/0$ case.

A first quantity to verify. If every response in a group gets the same reward, all advantages are zero and the group gives no gradient. With per-prompt success p, the probability that a group is informative is

$$\phi(p, G) = 1 - p^G - (1 - p)^G \approx Gp \quad (p \ll 1). \tag{3}$$

Equation (3) links all three knobs: sparsity sets p, the initial success rate is p at step 0, and the group size is G. Because our p is exact, matching Eq. (3) validates the harness; the experiments ask how this per-step quantity translates into learning speed over a run.

3.1 Success amplification at finite group size

Mroueh [2025] works in the limit where the group statistics equal their population values. Practical runs use $G \in [2, 64]$, and at small p most groups are uninformative, so we extend the fixed point to finite G. Fix a prompt with success probability p under the current policy, and let $r_2, \ldots, r_G \sim \text{Bernoulli}(p)$ be the other rewards in a group. Define the expected advantage of a correct and of an incorrect response,

$$a_G^+(p) = \mathbb{E}[A_1 \mid r_1 = 1], \qquad a_G^-(p) = -\mathbb{E}[A_1 \mid r_1 = 0], \qquad \Omega_G(p) = a_G^+(p) + a_G^-(p). \tag{4}$$

Both are finite sums over $j \sim \text{Bin}(G-1, p)$, the number of other correct responses. For `grpo`,

$$\Omega_G(p) = \sum_{j=0}^{G-1} \binom{G-1}{j} p^j (1-p)^{G-1-j} \left[\frac{1 - \frac{j+1}{G}}{\sqrt{\frac{j+1}{G}\left(1 - \frac{j+1}{G}\right) + \epsilon}} + \frac{\frac{j}{G}}{\sqrt{\frac{j}{G}\left(1 - \frac{j}{G}\right) + \epsilon}} \right]. \tag{5}$$

Proposition 1. *Let the policy be a softmax over all responses with independent logits for each prompt (e.g. a tabular policy with $L = 1$), and let updates be on-policy with the exact KL penalty. Any stationary point of the expected update satisfies*

$$\operatorname{logit} p^* = \operatorname{logit} p_{\text{ref}} + \frac{\Omega_G(p^*)}{\beta}. \tag{6}$$

In particular, $\Omega_G \equiv (G-1)/G$ for `mean-only`, $\Omega_G \equiv 1$ for `rloo`, and $\Omega_2 \approx 1$ for `grpo`, so these three fixed points are explicit. As $G \to \infty$, $\Omega_G(p) \to (p(1-p) + \epsilon)^{-1/2}$ and Eq. (6) recovers the fixed-point equation $p^* = h(p^*)$ of Mroueh [2025].

The proof (Section A) uses exchangeability of the group to write the expected update as the gradient of a reward-tilted objective, whose KL-regularized optimum is $\pi \propto \pi_{\text{ref}} \exp(w(r)/\beta)$. The stationary points of on-policy SGD are therefore exactly the fixed points of Mroueh's exact iteration, with Ω_∞ replaced by Ω_G. A Monte Carlo check over $G \in \{2, 4, 16\}$ and $p \in \{0.05, 0.3, 0.7\}$ agrees with Eq. (5) and the closed forms to within 0.01.

The proposition also connects to the view of Davis and Recht [2025]. For the same policy class, maximizing $F(p) - \beta\,\mathrm{KL}(\pi \,\|\, \pi_{\text{ref}})$ for a monotone F gives the stationarity condition $\operatorname{logit} p = \operatorname{logit} p_{\text{ref}} + F'(p)/\beta$. Our fixed points are therefore those of an effective objective F_G with $F_G' = \Omega_G$: $F_G(p) = \frac{G-1}{G}\,p$ for `mean-only`, $F_G(p) = p$ for `rloo`, and, for `grpo` as $G \to \infty$ with $\epsilon = 0$, $F_\infty(p) = 2 \arcsin \sqrt{p}$, whose derivative is $1/\sqrt{p(1-p)}$. This matches Davis and Recht's transform. At the success rates in Table 1, finite G replaces it with a flatter one.

Same fixed points, different stability. Let $z = \operatorname{logit} p$. For the tabular policy with $L = 1$, the wrong answers start with equal logits and stay equal, so the expected-gradient flow reduces to a scalar equation (Section A):

$$\dot{z} = \kappa(z)\Big[\Omega_G(\sigma(z)) - \beta\,(z - \operatorname{logit} p_{\text{ref}})\Big], \qquad \kappa(z) > 0. \tag{7}$$

A fixed point is stable under this flow when its slope $s = \Omega_G'(p^*)\,p^*(1-p^*)/\beta$ satisfies $s < 1$. Under Mroueh's map it is stable only when $|s| < 1$. Starting from p_{ref}, the flow climbs to the smallest fixed point above p_{ref}, which always has $s < 1$ and is therefore stable. The exact map can instead overshoot it. With $G = \infty$ and $p_{\text{ref}} = 10^{-3}$, our numerical solution of the map gives a 2-cycle for $\beta \in [3, 5]$, consistent with Mroueh's reported divergence. For $\beta \in [0.5, 2]$ the map escapes to $p \approx 1$,

Table 1: Effective GRPO weight $\Omega_G(p)$ from Eq. (5) ($\epsilon = 10^{-5}$). The $G = \infty$ column is the value used by Mroueh [2025].

p	$G = 2$	$G = 4$	$G = 16$	$G = \infty$	`mean-only`, $G = 16$
0.05	1.00	1.71	3.27	4.59	0.94
0.30	1.00	1.63	2.09	2.18	0.94

while the flow settles at $p^* \in [0.02, 0.17]$. With $p_{\text{ref}} = 1/8$, both converge to the same point for every β in our grid, so this effect only appears for sparse rewards.

Table 1 shows that the correction is large where it matters: at $p = 0.05$ and $G = 16$, the finite-G weight is 29% below the infinite-G value, so the infinite-G formula overstates how much std-normalization amplifies sparse prompts. The proposition also makes the `grpo` vs. `mean-only` comparison precise. At equal β, `grpo`'s larger Ω_G acts largely as a smaller effective β, so we compare the two both at equal β and at matched effective strength, to separate scale from shape. The proposition does not cover the factorized policy with $L > 1$, which cannot represent a tilt on the joint event "all positions correct". Measuring that gap is part of E2.

Metrics. Exact pass@1 and pass@k on held-out prompts; the measured fraction of informative groups, compared with the batch average $\mathbb{E}_x[\phi(p(x), G)]$ of Eq. (3) and with $\phi(\mathbb{E}_x[p(x)], G)$, which overestimates it when $p(x)$ varies across prompts [Nie et al., 2026]; diversity, measured as policy entropy, the number of distinct responses per group, and (in the multi-solution variant) the mass on the rarest correct responses; the log-probability–advantage covariance that drives entropy change [Cui et al., 2025]; exact KL to π_{ref}; gradient norm; and steps to reach a target success rate. We report a run as *stalled* if it never improves on p_0, and as *collapsed* if its final success falls below p_0. All results are averaged over at least 3 seeds.

4 Plan

Ambitious goal. Produce a map of when GRPO learns, stalls, or collapses as a function of $(p_0, G, \text{sparsity}, \text{step size})$, with each region explained by one of the theory papers, and confirm the main trends on a small LLM.

Core experiments (what we commit to).

E1 Initial success rate × group size. Sweep $L \in \{1, \ldots, 5\}$ (p_0 from 0.25 to 10^{-3}) and $G \in \{2, 4, 8, 16, 32\}$. We run two budgets: a fixed number of prompts per step, and a fixed number of samples per step ($B \times G$ constant). Hypotheses: the measured informative-group fraction follows Eq. (3) (a harness check); a larger G helps mostly when p_0 is small; at a fixed sample budget, the advantage of larger G shrinks; entropy and distinct responses fall as success rises. At a fixed sample budget we expect $G = 2$ to match large G when p_0 is moderate [Wu et al., 2025] but to fall behind when p_0 is small, where very large groups are needed [Hu et al., 2025]; locating the crossover in p_0 reconciles the two results. We also test the prediction of Ghosh et al. [2026] that the mean trajectory is nearly invariant to G, and compare our measured learning speeds and gradient variances with the group-size laws of Bay and Yearick [2026] and Zhou et al. [2026]. Finally, we repeat the grid with prompts whose p_0 is spread out rather than equal, where Ge et al. [2026] predict that std-normalization helps most and Nie et al. [2026] predict more all-equal groups than the average p_0 suggests.

E2 Success amplification [Mroueh, 2025]. With $L = 1$, use $K = 8$ ($p_{\text{ref}} = 1/8$) and $K = 1000$ ($p_{\text{ref}} = 10^{-3}$), and sweep $\beta \in \{0.1, 0.25, 0.5, 1, 2, 4, 8\}$ and $G \in \{2, 4, 16\}$ for `grpo`, `mean-only` and `rloo`, with plain SGD. We solve Eq. (6) numerically for each setting and run each configuration with both a tabular policy, where Proposition 1 applies exactly, and the MLP. Hypotheses: (i) `mean-only` and `rloo` converge to their explicit fixed points, which validates the harness; (ii) `grpo` converges to the finite-G fixed point p_G^* rather than to Mroueh's $G \to \infty$ value, with the gap largest at small G and small p_{ref}; (iii) at $G = 2$, `grpo` and `rloo` reach the same success rate; (iv) at $p_{\text{ref}} = 10^{-3}$, small-step SGD settles at the smallest fixed point even where Mroueh's map oscillates or escapes to $p \approx 1$, and raising the learning rate moves SGD toward the map's behaviour, which links E2 to E3. The gap between the tabular policy and the MLP measures the effect of function approximation. We repeat (ii) with $L = 2$ to measure the factorization gap. Because we compute the KL exactly, we can also measure how much of each update comes from the KL term on groups with all-equal rewards, one of the failure modes listed by Ding [2026].

E3 Step-size threshold [Suk and Duan, 2025]. Start every run from the same p_0 and sweep the SGD learning rate over a log grid. Without a KL penalty, the `grpo`, `mean-only` and `rloo` advantages sum to zero within a group, so one step can only raise the logits of the correct tokens and lower those of sampled wrong ones. On the single-solution task, a policy with separate logits for each prompt and position therefore cannot lose success at any learning rate, and any threshold must come from parameters shared across prompts and positions. E3 therefore uses the MLP, with the tabular policy as a control that should show no threshold. *Fixed length:* for $L \in \{1, 2, 4, 8\}$ with summed log-probabilities, we record the largest learning rate $\bar{\eta}(L)$ that still improves on p_0. With a fixed L, averaging the log-probabilities over tokens only rescales the learning rate by $1/L$, so this arm tests the predicted $\bar{\eta} \propto 1/L$ scaling rather than comparing normalizations. *Mixed length:* prompts with $L(x) \in \{2, \ldots, 8\}$ share one learning rate, and we sweep the length weighting $f_\alpha(L) = L^{\alpha-1}$ of Ding et al. [2026] over $\alpha \in \{0, 0.5, 1\}$, from per-response $1/|y|$ averaging (GRPO, $\alpha = 0$) to summed log-probabilities ($\alpha = 1$). Dr. GRPO's constant $1/L_{\max}$ normalizer equals the summed loss with the learning rate divided by $L_{\max}$, so the learning-rate sweep covers it. We run the sweep for both `grpo` and `mean-only`, since Ge et al. [2026] predict that std-normalization changes the effective step size. Hypotheses: there is a narrow learning-rate range where success first rises and then stalls or collapses; $\bar{\eta}$ shrinks roughly as $1/L$ and as p_0 grows (through $1 - J$); in the mixed-length task, the summed loss has its threshold set by the longest prompts, so short prompts learn slowly at any stable learning rate, while per-response averaging equalizes the thresholds across lengths. Within a prompt, all responses have the same length, so the response-length bias that Dr. GRPO targets is absent and only the step-size effect remains.

Extensions (if time permits).

E4 Diversity under pass@k shaping [Thrampoulidis et al., 2026]. On the multi-solution task, compare `grpo` with pass@k advantages ($k = 4, 8$, with $G = 16 \geq k$), the focal-style weighting of F-GRPO [Plyusov et al., 2026], and updating only on prompts with no observed success [Yuan et al., 2026]. We expect higher entropy and pass@16 at some cost to pass@1, and test three further predictions: a non-monotonic effect of G on the mass of rare correct responses [Plyusov et al., 2026]; the entropy–performance law of Cui et al. [2025], whose driving covariance we can compute exactly; and the crossover of Yue et al. [2025], where the trained policy overtakes the base model at small k and falls behind it at large k.

E5 Escaping very sparse rewards. At $L = 6$, $p_0 = 4^{-6} \approx 2.4 \times 10^{-4}$ (where Eq. (3) predicts almost no signal), compare a very large G [Hu et al., 2025], mixing in easier prompts, a curriculum over L, sampling prompts in proportion to $p(1-p)$ [Foster et al., 2025] or to $\phi(p, G)$ [cf. Sha et al., 2026; Okonkwo et al., 2026], and the fixed-reference advantage $A = 2r - 1$, which still learns from all-wrong groups [Nie et al., 2026]. In our task the exact p lets us run the prompt-selection arms with the true difficulty, an upper bound on what an estimated difficulty can achieve.

E6 Small-LLM check. We first reproduce the experiment of Mroueh [2025, App. A], which uses only public components: Qwen2.5-0.5B-Instruct [Qwen Team, 2025] on GSM8K [Cobbe et al., 2021], trained with TRL's GRPOTrainer [von Werra et al., 2020] and vLLM, $G = 16$, learning rate 5×10^{-6}, $\beta = 0.1$, a 200-token context, and sampling temperature 0.1. Its reported success rises from 21% to 37.5%. We then repeat E1 on the LLM. We estimate each training prompt's success rate from 16 samples of the base model, bucket prompts at $p \approx \{0.05, 0.2, 0.5, 0.8\}$, and run $G \in \{2, 4, 8, 16\}$ within each bucket (2 seeds, about 200 steps), logging the informative-group fraction against Eq. (3). We add a random-reward control run [Shao et al., 2025] and plot pass@k of the base and trained models up to $k = 256$ to check for the crossover of Yue et al. [2025]; Tan et al. [2025] report reference curves for Qwen2.5-0.5B. `mean-only` is TRL's `scale_rewards=False`; for E4 we use verl's built-in pass@k advantage or a small subclass of the TRL trainer. If GSM8K has too few prompts in the hard buckets, we use MATH [Hendrycks et al., 2021] difficulty levels 1–5 instead. Runs use 40 GB MIG slices of H100 GPUs on the DRAC Fir cluster. We will time a pilot run before committing; at an assumed $\sim$2 GPU-hours per run, the 32-run grid costs about 3 GPU-days. We drop the multiple-choice variant of our earlier plan because its chance level ($\geq 25\%$) cannot reach the sparse regime.

Current status. We have derived Proposition 1 and computed the predicted fixed points and their stability (Table 1, Eq. (7)). The training harness (task, policy, advantage estimators, and exact metrics) is not yet implemented; it is the first milestone.

Timeline and roles. Weeks 1–2: build the harness and unit-test the exact metrics against Eq. (3) and the closed-form fixed points; complete E1. Weeks 3–4 (milestone): complete E2 and E3, and set up the E6 pilot run. Weeks 5–6: extensions E4–E6, ablations, and the final report. Each member owns one experiment end-to-end (implementation, runs, figures, and the related section of the report), and all four share the code and the literature review.

What we expect to learn. The theory is derived for exact gradients and simple policies. Where our curves deviate from it, we will identify the cause: finite G (which Proposition 1 quantifies for E2), stochastic batches, function approximation shared across prompts, or the factorized multi-token policy. These deviations are the most useful output of the project.

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A Proof of Proposition 1

Fix a prompt and write $\pi = \pi_\theta(\cdot \mid x)$ and $p = \sum_{y:r(y)=1} \pi(y)$. The advantages are computed from sampled rewards and are not differentiated, so the expected update direction of the reward term is $g = \mathbb{E}[\frac{1}{G}\sum_{i=1}^{G} A_i \nabla_\theta \log \pi(y_i)]$. The group is exchangeable, so $g = \mathbb{E}[A_1 \nabla_\theta \log \pi(y_1)]$. Conditional on y_1, the other responses are independent of y_1, so $\mathbb{E}[A_1 \mid y_1]$ depends on y_1 only through $r(y_1)$, and equals $w(1) = a_G^+(p)$ or $w(0) = -a_G^-(p)$. Hence

$$g = \sum_y \pi(y)\, w(r(y))\, \nabla_\theta \log \pi(y) = \nabla_\theta \sum_y \pi_\theta(y)\, w(r(y))\Big|_{w \text{ fixed}}.$$

Adding the exact KL penalty, a stationary point is a stationary point of $F(\pi) = \sum_y \pi(y) w(r(y)) - \beta\, \mathrm{KL}(\pi \,\|\, \pi_{\mathrm{ref}})$ at fixed w. When the policy class contains every distribution with full support (a softmax over all responses), $\nabla_\theta F = 0$ if and only if $w(r(y)) - \beta(\log \frac{\pi(y)}{\pi_{\mathrm{ref}}(y)} + 1)$ is constant in y, i.e. $\pi(y) \propto \pi_{\mathrm{ref}}(y) \exp(w(r(y))/\beta)$. Summing over correct and incorrect responses,

$$\frac{p}{1-p} = \frac{p_{\mathrm{ref}}}{1-p_{\mathrm{ref}}} \exp\Big(\frac{a_G^+(p) + a_G^-(p)}{\beta}\Big),$$

which is Eq. (6), with p evaluated at the stationary point itself.

For `mean-only`, $A_1 = r_1 - \frac{1}{G}\sum_i r_i$, so $a_G^+ = 1 - \frac{1+(G-1)p}{G} = \frac{G-1}{G}(1-p)$ and $a_G^- = \frac{G-1}{G}p$, giving $\Omega_G = (G-1)/G$. For `rloo`, $A_1 = r_1 - \frac{1}{G-1}\sum_{i\geq 2} r_i$, so $a_G^+ = 1-p$, $a_G^- = p$ and $\Omega_G = 1$. For `grpo` with $G = 2$, the only informative group has rewards $\{0,1\}$ and advantages $\pm\frac{1}{2}/\sqrt{\frac{1}{4}+\epsilon} \approx \pm 1$, so $a_2^+ \approx 1-p$, $a_2^- \approx p$ and $\Omega_2 \approx 1$. As $G \to \infty$, $j/G \to p$ in probability, so $a_G^+ \to (1-p)/\sqrt{p(1-p)+\epsilon}$ and $a_G^- \to p/\sqrt{p(1-p)+\epsilon}$ by bounded convergence, and $\Omega_G \to (p(1-p)+\epsilon)^{-1/2}$, the weight in the recurrence of Mroueh [2025]. If the implementation uses the Bessel-corrected standard deviation, replace $\frac{c}{G}(1-\frac{c}{G})$ in Eq. (5) by $\frac{c(G-c)}{G(G-1)}$. □

Derivation of Eq. (7). Take $L = 1$, K answers with one correct answer c, and logits ℓ_k, so $\pi_k \propto e^{\ell_k}$. Write $g_k = w(r_k) - \beta(\log \frac{\pi_k}{\pi_{\mathrm{ref},k}} + 1)$ and $\bar{g} = \sum_k \pi_k g_k$. The expected-gradient flow on F is $\dot{\ell}_k = \pi_k(g_k - \bar{g})$. With a zero-initialized head, the $K-1$ wrong answers start with equal logits and receive equal updates, so they stay equal, each with probability $\pi_w = (1-p)/(K-1)$. Let $\Delta = g_c - g_w$. Then $\dot{\ell}_c = p(1-p)\Delta$ and $\dot{\ell}_w = -p\,\pi_w \Delta$. Using $\pi_{\mathrm{ref},w} = (1-p_{\mathrm{ref}})/(K-1)$, $\Delta = a_G^+ + a_G^- - \beta(z - \mathrm{logit}\, p_{\mathrm{ref}})$, where $z = \mathrm{logit}\, p = \ell_c - \ell_w - \log(K-1)$. Hence $\dot{z} = \dot{\ell}_c - \dot{\ell}_w = \kappa\,\Delta$ with $\kappa = p(1-p) + p\,\pi_w > 0$, which is Eq. (7). A fixed point is stable when $\partial\Delta/\partial z < 0$, i.e. when

$\Omega'_G(p^*)\,p^*(1-p^*) < \beta$. Mroueh's map $z_n = \operatorname{logit} p_{\text{ref}} + \Omega(\sigma(z_{n-1}))/\beta$ has slope $s = \Omega'(p^*)\,p^*(1-p^*)/\beta$ at a fixed point and is stable when $|s| < 1$. The two conditions agree when $s > 0$; when $s < -1$, the map oscillates but the flow is still stable.

B Experimental Details

Component	Setting
Prompt features	$x \sim \mathcal{N}(0, I_{16})$; 512 train / 256 held-out test prompts
Task	$K = 4$ (E1, E3–E5), $K \in \{8, 1000\}$ (E2); $L \in \{1, \ldots, 8\}$; Gaussian W_t
Mixed-length task (E3)	$L(x)$ uniform on $\{2, \ldots, 8\}$, fixed per prompt
Multi-solution task (E4)	m of K tokens correct per position; unequal initial probabilities on correct tokens
Policy	MLP (2 hidden layers, width 128) on (x, one-hot t); zero-init head, so π_{ref} is uniform. E2 and E3 also use a tabular softmax per prompt
Optimizer	Adam (E1, E4, E5); plain SGD for E2 and the learning-rate sweep (E3)
Batch	$B = 32$ prompts $\times$ G samples; fixed-compute runs use $B \times G = 256$
Advantage std	$\sqrt{\bar{r}(1-\bar{r}) + \epsilon}$, $\epsilon = 10^{-5}$, no Bessel correction
Seeds	3 per configuration
Compute	CPU only; each run takes seconds to minutes
LLM (E6)	Qwen2.5-0.5B-Instruct; GSM8K (MATH levels 1–5 as fallback)
LLM training	TRL GRPOTrainer + vLLM; $G \leq 16$, lr 5×10^{-6}, $\beta = 0.1$, 200-token context, temperature 0.1
LLM compute	H100 MIG slices (40 GB) on DRAC Fir; ≈3 GPU-days budget